



How to Run Ansible Playbook Locally | Devops Junction

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In this post, we are going to see how to run the playbook locally on our Localhost on the control machine. How to run Ansible Playbook locally is one of the questions I used to have before I learnt it.

Let me tell you how to Run Ansible Playbook Locally with Examples.

In fact, There are many ways to run ansible playbooks locally. Let us see each one and how they work.

How to run Ansible Playbook Locally on Control Machine

www.middlewareinventory.com

There are four ways to run ansible playbook locally and we have listed them all here.

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Method1: Specify Localhost in your hosts directive of your playbook.

If you are running a playbook which you want to run on localhost (or) in other words you have a playbook you want to run locally. Within the playbook Just mention `localhost` in the hosts segment where you usually specify the host group. Consider the following playbook.

I have just specified the `localhost` in the hosts directive or key.

When you run this playbook. Despite you have no entry named localhost in your ansible inventory file `/etc/ansible/hosts` or custom hosts file specified with `-i` option. It would work

As an additional measure that it should run in `localhost` or locally. You can add a one more parameter named `connection` and set it to local

```
---
- name: "Playing with Ansible and Git"
  hosts: localhost
  connection: local
  tasks:

    - name: "just execute a ls -lrt command"
      shell: "ls -lrt"
      register: "output"

    - debug: var=output.stdout_lines
```

Here is the Test Results Snapshot of my Terminal.

```
→ Ansible-Examples git:(master) x pwd
/apps/vagrant/webinfra/AnsibleWorkspace/Ansible-Examples
→ Ansible-Examples git:(master) x ls -lrt
total 8
-rw-r--r-- 1 aksarav wheel 212 Jul 17 13:50 Ansible-Local.yml
-rw-r--r-- 1 aksarav wheel  0 Jul 17 13:59 sometestfile
-rw-r--r-- 1 aksarav wheel  0 Jul 17 13:59 sometestfile1
→ Ansible-Examples git:(master) x ansible-playbook Ansible-Local.yml

PLAY [Playing with Ansible and Git] *****

TASK [Gathering Facts] *****
ok: [localhost]

TASK [just execute a ls -lrt command] *****
changed: [localhost]

TASK [debug] *****
ok: [localhost] => {
  "output.stdout_lines": [
    "total 8",
    "-rw-r--r-- 1 aksarav wheel 212 Jul 17 13:50 Ansible-Local.yml",
    "-rw-r--r-- 1 aksarav wheel  0 Jul 17 13:59 sometestfile",
    "-rw-r--r-- 1 aksarav wheel  0 Jul 17 13:59 sometestfile1"
  ]
}

PLAY RECAP *****
localhost : ok=3  changed=1  unreachable=0  failed=0

→ Ansible-Examples git:(master) x
```

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Method2: Using `local_action` clause in the ansible playbook

This method is most recommended than the other 3 but it is always subjective to use it for your requirements.

Here is the example playbook that runs locally to create an EC2 instance connecting to AWS.

if you notice, we have not specified any `hosts` in here alternatively we have used `local_action` clause to specify that this task has to run local.

```
- name: create an ec2 instance
  local_action:
    module: ec2
    image: ami-8caa1ce4
    instance_type: m3.medium
    key_name: mykey
    group:
      - web
      - ssh
    instance_tags:
      type: web
      env: production
```

here is the fully-functioning example of this method.



Ansible EC2 Example - Create EC2 instance with Ansible

0 Infrastructure as Code is getting all attention it deserves and everyone is trying to find their way to the `Completely automated Infrastructure Provisioning & Management` While there are a lot of tools available now in the market starting from Terraform, AWS CloudFormation, Chef, Puppet, Salt Stack There are some ... [Continue reading](#)

 **Middleware Inventory** 0 

Method3: Add an entry in your Inventory

This is a second method to run ansible playbook locally.

You can also explicitly define your localhost in your inventory file. your inventory file can be at the default location `/etc/ansible/hosts` or a customized `ansible_hosts` file in

your present working directory.

```
→ Ansible-Examples git:(master) ✕ cat ansible_hosts Local Ansible_hosts file / Inventory
[app]
mwivmapp01
mwivmapp02

localhost ansible_connection=local
→ Ansible-Examples git:(master) ✕ cat /etc/ansible/hosts Default Inventory
[app]
mwivmapp01
mwivmapp02

localhost ansible_connection=local
→ Ansible-Examples git:(master) ✕
```

Now if you run the same playbook we have given above you would yield the same results.

In the preceding Snapshot you can see the Yellow highlighted line were added to the inventory file specifying the localhost.

In this way if you would like to add your localhost with someother alias name. like "controlmachine" you can do the same.

```
→ Ansible-Examples git:(master) ✕ ansible controlmachine -m ping -i ansible_hosts
controlmachine | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
→ Ansible-Examples git:(master) ✕ ansible controlmachine -m shell -a "ls -lrt" -i ansible_hosts
controlmachine | CHANGED | rc=0 >>
total 16
-rw-r--r-- 1 aksarav wheel  0 Jul 17 13:59 sometestfile
-rw-r--r-- 1 aksarav wheel  0 Jul 17 13:59 sometestfile1
-rw-r--r-- 1 aksarav wheel 206 Jul 17 17:14 Ansible-Local.yml
-rw-r--r-- 1 aksarav wheel  69 Jul 17 17:21 ansible_hosts

→ Ansible-Examples git:(master) ✕ cat ansible_hosts
[app]
mwivmapp01
mwivmapp02

controlmachine ansible_connection=local
→ Ansible-Examples git:(master) ✕
```

Now you can refer your localhost by this name into your playbook like `hosts: controlmachine`

Method4: Specify in the Ansible Command line

This is a third method to run ansible playbook locally.

By Default, Ansible would run the playbook on the host group which is mentioned in the playbook with `hosts:` directive.

But if you want to ignore all those hosts specified in the playbook and run it locally. You can use this method.

In this way. The hosts directive would be ignored and your task would run only on localhost.

I have a host group named `app` in my Ansible inventory file, it has two hosts beneath it and the playbook is defined to run on all the hosts in my inventory with `hosts: all`

```

→ Ansible-Examples git:(master) ✕ cat ansible_hosts ## MY INVENTORY FILE
[app]
mwivmapp01
mwivmapp02

controlmachine ansible_connection=local
→ Ansible-Examples git:(master) ✕ cat Ansible-Local.yml ## MY PLAYBOOK
---
- name: "Playing with Ansible and Git"
  hosts: all
  tasks:

    - name: "just execute a ls -lrt command"
      shell: "ls -lrt"
      register: "output"

    - debug: var=output.stdout_lines

→ Ansible-Examples git:(master) ✕

```

When I run this playbook. the Playbook would be run against all the host groups specified in my inventory file.

For some reason, I don't want to run this playbook on any of remote hosts yet. I want to test it locally and run it locally. What should I do?

All you should do is this

```

ansible-playbook \
--connection=local \
--inventory 127.0.0.1, \
--limit 127.0.0.1 Ansible-Local.yml -i ansible_hosts

```

Here

- `--connection` - tells ansible to run the file locally.

- `--inventory` - tells ansible to consider this host name as an inventory [Comma at the end is IMPORTANT]

- `--limit` - Limiting to only this host.

```

→ Ansible-Examples git:(master) ✕ ansible-playbook --connection=local --inventory 127.0.0.1, --limit 127.0.0.1 Ansible-Local.yml -i ansible_hosts

PLAY [Playing with Ansible and Git] *****

TASK [Gathering Facts] *****
ok: [127.0.0.1]

TASK [just execute a ls -lrt command] *****
changed: [127.0.0.1]

TASK [debug] *****
ok: [127.0.0.1] => {
  "output.stdout_lines": [
    "total 16",
    "-rw-r--r-- 1 aksarav wheel  0 Jul 17 13:59 sometestfile",
    "-rw-r--r-- 1 aksarav wheel  0 Jul 17 13:59 sometestfile1",
    "-rw-r--r-- 1 aksarav wheel 29 Jul 17 14:48 ansible_hosts",
    "-rw-r--r-- 1 aksarav wheel 206 Jul 17 17:14 Ansible-Local.yml"
  ]
}

PLAY RECAP *****
127.0.0.1          : ok=3    changed=1    unreachable=0    failed=0

```

Why `--limit` is important here in method3

Note*: The Same command can be used without `--limit` but it would consider all the hosts mentioned in the hosts directive of the playbook.

Though it considers the remote servers it would not be able to run any task on those remote servers as we have set the conneciton to local.

Here is the confusing part.

What would ansible do here is that. It would execute the command locally but it will print the results in the name of remote servers.

```
→ Ansible-Examples git:(master) x ansible-playbook --connection=local --inventory 127.0.0.1, Ansible-Local.yml -i ansible_hosts

PLAY [Playing with Ansible and Git] *****

TASK [Gathering Facts] *****
ok: [controlmachine]
ok: [127.0.0.1]
ok: [mwivmapp02] ← Look All the Hosts are considered
ok: [mwivmapp01]

TASK [just execute a ls -lrt command] *****
changed: [127.0.0.1]
changed: [controlmachine]
changed: [mwivmapp01]
changed: [mwivmapp02]

TASK [debug] *****
ok: [127.0.0.1] => {
  "output.stdout_lines": [
    "total 16",
    "-rw-r--r-- 1 aksarav wheel  0 Jul 17 13:59 sometestfile",
    "-rw-r--r-- 1 aksarav wheel  0 Jul 17 13:59 sometestfile1",
    "-rw-r--r-- 1 aksarav wheel 206 Jul 17 17:14 Ansible-Local.yml",
    "-rw-r--r-- 1 aksarav wheel 69 Jul 17 17:21 ansible_hosts"
  ]
}
ok: [mwivmapp01] => {
  "output.stdout_lines": [
    "total 16",
    "-rw-r--r-- 1 aksarav wheel  0 Jul 17 13:59 sometestfile",
    "-rw-r--r-- 1 aksarav wheel  0 Jul 17 13:59 sometestfile1",
    "-rw-r--r-- 1 aksarav wheel 206 Jul 17 17:14 Ansible-Local.yml",
    "-rw-r--r-- 1 aksarav wheel 69 Jul 17 17:21 ansible_hosts"
  ]
}
ok: [mwivmapp02] => {
  "output.stdout_lines": [
    "total 16",
    "-rw-r--r-- 1 aksarav wheel  0 Jul 17 13:59 sometestfile",
```

Thanks,
Sarav AK



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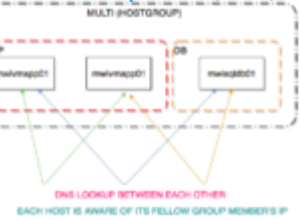
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ansible update /etc/hosts file with IP of all hosts across all hosts



In Ansible, SSH Communication is everything. Let it be connecting to remote servers from your controller laptop/desktop (mac/windows) or to enable connectivity between servers on the host group. Especially this will be a huge task to accomplish when you do not have a DNS where each remote host can easily...



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How to Copy files between remote hosts in ansible

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ansible search for string in file or check if string exists in file

The Objective of this post is to show how to search for a string in a file with ansible. ansible provides various ways to accomplish the same. We will cover, three major ways to search for a string in a file. Lineinfile module Using the Shell module and grep command...

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